

Customer Retention Prediction — Auto Analysis

Business Analytics Report

Project: Customer Retention Prediction

Dataset: customer_retention_training.csv

Analysis date: 2026-09-24

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Generated by the AI Data Scientist Platform

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1. Executive Summary

This report analyzes **customer_retention_training.csv** (3,000 rows, 22 columns) covering 2024-01-13 to 2026-05-31 .

Key verified findings

- The original dataset contained 2,273 missing value(s) and 25 duplicate row(s), which have since been cleaned.
- A Gradient Boosting model was trained to predict **Will Return Next 90 Days** (classification). The model correctly classifies about 66.7% of cases (Accuracy).
- 2 features excluded as data leakage before training (Coupon Usage Ratio, Orders Next 90 Days) to avoid an artificially inflated score.

Model reliability, in plain terms: The model correctly classifies about 66.7% of cases (Accuracy). F1 score (balancing precision and recall) is 0.667. ROC-AUC is 0.716 (1.0 = perfect separation, 0.5 = no better than chance).

Top recommendations

- The most common observed outcome for Will Return Next 90 Days is 'No' (334 of 600 test cases).
- Among records, the 'Days Since Last Order: Lower' group shows the highest observed rate (61.6%, n=1000), compared to 'Days Since Last Order: Higher' (20.3%, n=1000).

2. About the Dataset

customer_retention_training.csv contains 3,000 rows and 22 columns. It spans the period 2024-01-13 to 2026-05-31 (based on Signup Date).

Column dictionary

Column	Type	Missing	Notes
Customer Id	Identifier	0.0%	
Signup Date	Date/Time	0.0%	
City	Category	0.0%	most common: Karachi
Gender	Category	0.0%	most common: Male
Age	Numeric	0.0%	typical range 0.00–212.00
Acquisition Channel	Category	0.0%	most common: Facebook Ads
Membership Tier	Category	0.0%	most common: Silver
Preferred Payment	Category	0.0%	most common: Cash on Delivery
Uses Mobile App	Category	0.0%	most common: No
Newsletter Subscribed	Category	0.0%	most common: No
Total Orders	Numeric	0.0%	typical range 1.00–27.00
Total Spent Pkr	Numeric	0.0%	typical range 311.00–124,900.00
Avg Order Value Pkr	Numeric	0.0%	typical range 318.00–11,245.00

Orders Last 90 Days	Numeric	0.0%	typical range 0.00–8.00
Days Since Last Order	Numeric	0.0%	typical range 1.00–371.00
Avg Delivery Delay Days	Numeric	0.0%	typical range 0.00–9.90
Product Returns	Numeric	0.0%	typical range 0.00–5.00
Complaints Filed	Numeric	0.0%	typical range 0.00–5.00
Coupon Usage Ratio	Numeric	0.0%	typical range 0.00–0.85
Avg Rating Given	Numeric	0.0%	typical range 1.10–5.00
Orders Next 90 Days	Numeric	0.0%	typical range 0.00–5.00
Will Return Next 90 Days	Category	0.0%	most common: No

3. Data Quality & Cleaning

Original dataset quality (as uploaded)

Total rows	Missing cells	Duplicate rows	Issues detected
3,025	2,273	25	18

Duplicate Rows

Column	Finding	Affected
(whole dataset)	25 duplicate row(s) detected.	25

Missing Values

Column	Finding	Affected
age	'age' has 123 missing value(s) (4.1%).	123
membership_tier	'membership_tier' has 1788 missing value(s) (59.1%).	1788
preferred_payment	'preferred_payment' has 90 missing value(s) (3.0%).	90
avg_delivery_delay_days	'avg_delivery_delay_days' has 90 missing value(s) (3.0%).	90
avg_rating_given	'avg_rating_given' has 182 missing value(s) (6.0%).	182

Impossible Values

Column	Finding	Affected
age	'age' has 1 negative value(s) that are rare exceptions among otherwise non-negative values — likely a data-entry error, not real signal.	1

Outliers

Column	Finding	Affected
age	'age' has 2 statistical outlier(s) outside [-7.0, 89.0].	2
total_orders	'total_orders' has 70 statistical outlier(s) outside [-4.5, 15.5].	70

total_spent_pkr	'total_spent_pkr' has 172 statistical outlier(s) outside [-21692.0, 53180.0].	172
avg_order_value_pkr	'avg_order_value_pkr' has 79 statistical outlier(s) outside [-1111.0, 6905.0].	79
orders_last_90_days	'orders_last_90_days' has 206 statistical outlier(s) outside [-1.5, 2.5].	206
days_since_last_order	'days_since_last_order' has 144 statistical outlier(s) outside [-71.0, 161.0].	144
avg_delivery_delay_days	'avg_delivery_delay_days' has 128 statistical outlier(s) outside [-1.85, 4.95].	128
product_returns	'product_returns' has 69 statistical outlier(s) outside [-1.5, 2.5].	69
complaints_filed	'complaints_filed' has 83 statistical outlier(s) outside [-1.5, 2.5].	83
coupon_usage_ratio	'coupon_usage_ratio' has 20 statistical outlier(s) outside [-0.19, 0.74].	20
avg_rating_given	'avg_rating_given' has 24 statistical outlier(s) outside [2.5, 5.7].	24

Cleaning performed

Problem	Before	Action	After	Reason
Duplicate rows	25 duplicate row(s)	Removed duplicate rows	0 duplicate rows	Duplicate records would otherwise double-count the same transaction/customer.
Missing Age	123 missing	Filled using the column median	0 missing	Reduces the influence of extreme values compared to the mean.
Missing Membership Tier	1788 missing	Filled using the most frequent value (mode)	0 missing	The most representative value for a categorical field.
Missing Preferred Payment	90 missing	Filled using the most frequent value (mode)	0 missing	The most representative value for a categorical field.
Missing Avg Delivery Delay Days	90 missing	Filled using the column median	0 missing	Reduces the influence of extreme values compared to the mean.
Missing Avg Rating Given	182 missing	Filled using the column median	0 missing	Reduces the influence of extreme values compared to the mean.

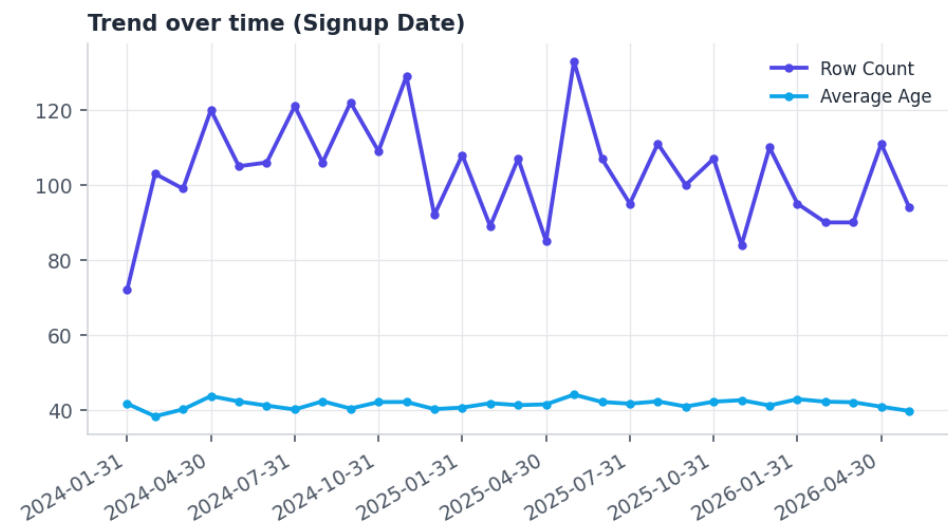
Impossible negative Age	1 impossible value(s)	Converted to missing, then filled	0 impossible values	A rare negative value in an otherwise non-negative column is almost always a data-entry error.
Inconsistent capitalization in City	6 variant label(s) (e.g. 'rainy' vs 'Rainy')	Merged into a single canonical label	consistent labels	Values differing only by letter case would otherwise be treated as different categories.

After cleaning: 0 missing cell(s), 0 duplicate row(s), 3,000 rows remain. The original file itself was never modified — cleaning produced a separate working copy.

4. Key EDA Findings

Total Rows	Will Return Next 90 Days Rate	Total Age	Total Total Orders
3,000	44.3%	124,549	17,819

Trend over time (Signup Date)





Orders Last 90 Days breakdown

Average: 0.76 (range 0.0–8.0)

Value	Count	Percent
8	1	0.0%
7	3	0.1%
6	9	0.3%
5	21	0.7%
4	32	1.1%
3	140	4.7%
2	365	12.2%
1	820	27.3%
0	1609	53.6%

5. Modeling Approach

A **classification** model was trained to predict **Will Return Next 90 Days**, using 17 feature(s).

Features used

Signup Date: Year, Signup Date: Month, Signup Date: Dayofweek, City, Gender, Age, Acquisition Channel, Membership Tier, Preferred Payment, Uses Mobile App, Newsletter Subscribed, Total Orders, Total Spent Pkr, Avg Order Value Pkr, Orders Last 90 Days, Days Since Last Order, Product Returns

Features excluded, and why

Column	Reason excluded
Customer Id	Identifier column (e.g. an ID) — carries no predictive signal.
Signup Date	Raw date — replaced with derived year / month / day-of-week features instead.
Coupon Usage Ratio	Leakage. Together with 'orders_next_90_days', this column almost exactly reproduces the target (100.0% agreement via a simple comparison) — the model could 'cheat' by combining the two instead of learning real patterns.
Orders Next 90 Days	Leakage. Together with 'coupon_usage_ratio', this column almost exactly reproduces the target (100.0% agreement via a simple comparison) — the model could 'cheat' by combining the two instead of learning real patterns.
Avg Delivery Delay Days	Excluded by default 'avg_delivery_delay_days' often represents information only available AFTER the outcome occurs (e.g. a rating or tip given once an order is complete), and shows a real (23% correlated) relationship with the target here. Review whether this would actually be known at prediction time — if not, exclude it below.
Complaints Filed	Excluded by default 'complaints_filed' often represents information only available AFTER the outcome occurs (e.g. a rating or tip given once an order is complete), and shows a real (18% correlated) relationship with the target here. Review whether this would actually be known at prediction time — if not, exclude it below.
Avg Rating Given	Excluded by default 'avg_rating_given' often represents information only available AFTER the outcome occurs (e.g. a rating or tip given once an order is complete), and shows a real (22% correlated) relationship with the target here. Review whether this would actually be known at prediction time — if not, exclude it below.

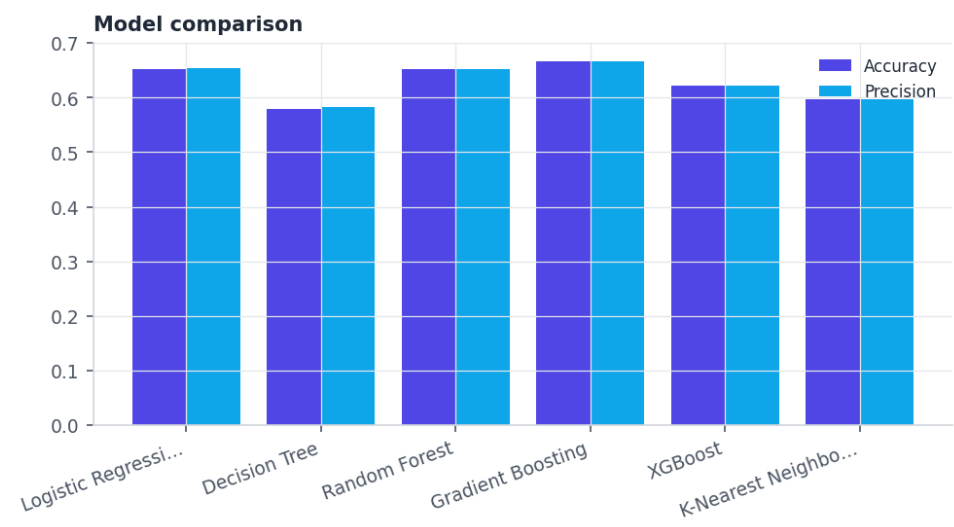
Technical details

Algorithm: Gradient Boosting · Preprocessing: 10 numeric + 7 categorical feature(s), scaled/encoded · Train/test split: 20.0% held out · Cross-validation folds: 5 · Random seed: 42

6. Model Performance

Model comparison

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.6533	0.6542	0.6533	0.6537	0.7199
Decision Tree	0.5800	0.5824	0.5800	0.5809	0.5772
Random Forest	0.6517	0.6527	0.6517	0.6521	0.7051
Gradient Boosting	0.6667	0.6675	0.6667	0.6670	0.7157
XGBoost	0.6217	0.6224	0.6217	0.6220	0.6801
K-Nearest Neighbors	0.5967	0.5964	0.5967	0.5965	0.6205

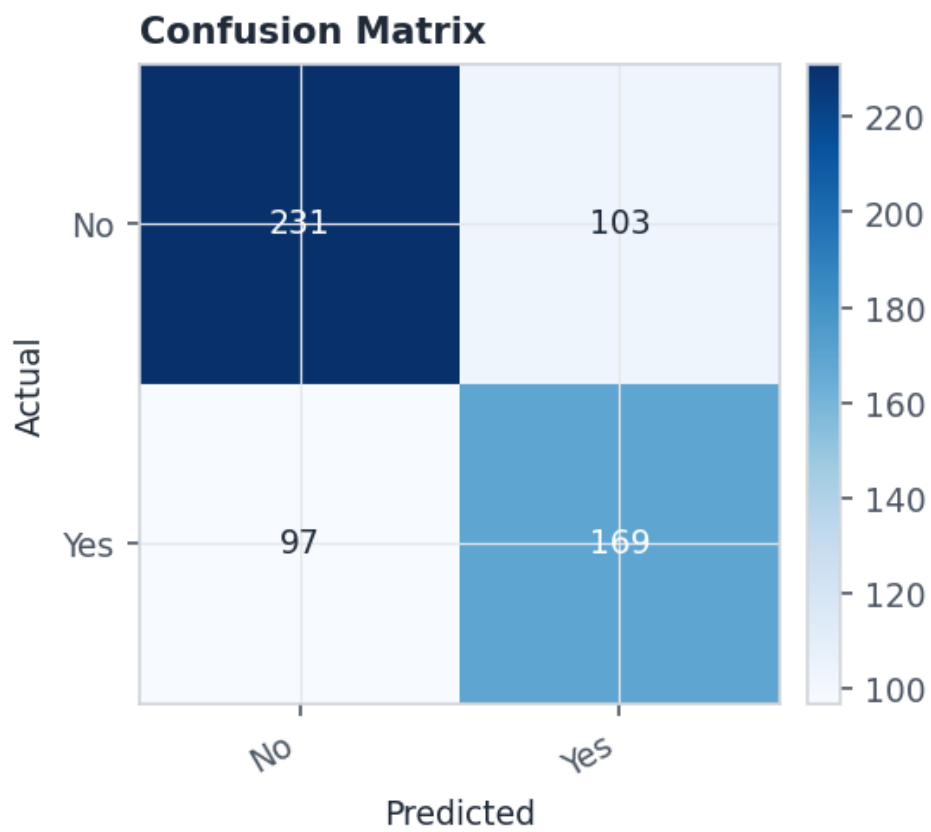


In plain language

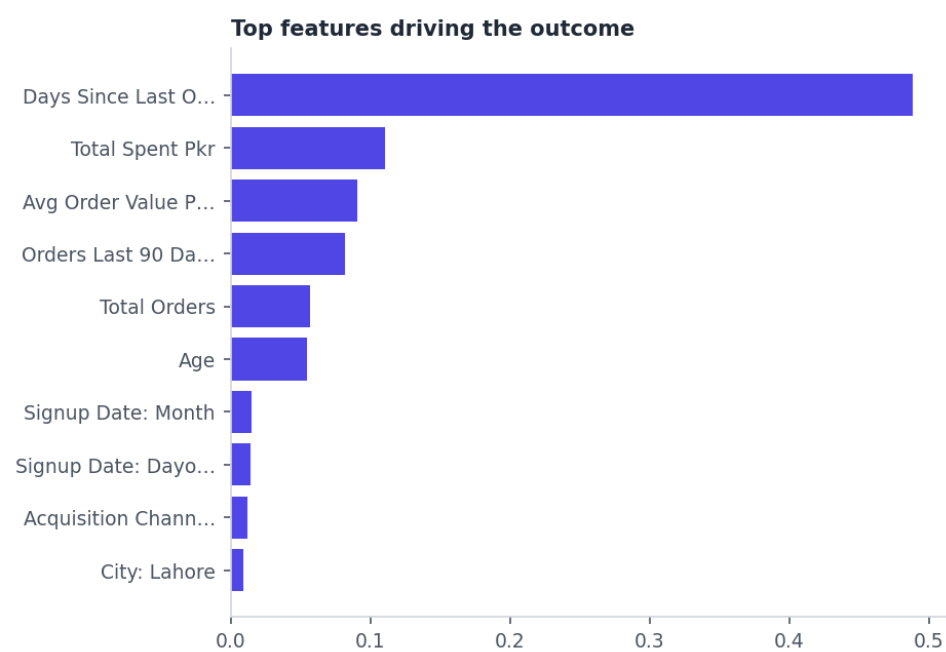
- The model correctly classifies about 66.7% of cases (Accuracy).
- F1 score (balancing precision and recall) is 0.667.
- ROC-AUC is 0.716 (1.0 = perfect separation, 0.5 = no better than chance).

Compared against a trivial baseline (F1 Score = 0.3981): this model clearly outperforms it.

Confusion Matrix



7. What Drives the Outcome



Feature	Relative importance
Days Since Last Order	48.9%
Total Spent Pkr	11.0%
Avg Order Value Pkr	9.1%
Orders Last 90 Days	8.2%
Total Orders	5.7%
Age	5.5%
Signup Date: Month	1.5%
Signup Date: Dayofweek	1.4%
Acquisition Channel: TikTok	1.2%
City: Lahore	0.9%

8. Future Outlook

- The most common observed outcome for Will Return Next 90 Days is 'No' (334 of 600 test cases).
- Among records, the 'Days Since Last Order: Lower' group shows the highest observed rate (61.6%, n=1000), compared to 'Days Since Last Order: Higher' (20.3%, n=1000).

Group	Observed rate	Group size
Days Since Last Order: Lower	61.6%	1000
Days Since Last Order: Middle	51.1%	1000
Days Since Last Order: Higher	20.3%	1000

9. AI Business Insights AI-generated

AI insights are unavailable right now (gemini: Gemini rate limit exceeded — too many requests. Try again shortly.; ollama fallback: The AI service is currently unavailable.). The rest of this report contains verified analytical results only.

10. Limitations

- The following column(s) were excluded as likely data leakage and could not be used: coupon_usage_ratio, orders_next_90_days. If any of these should genuinely be available before prediction time, review them manually.
- Cross-validation scores varied by ± 0.024 across folds — treat the headline metric as an estimate with this much natural uncertainty, not an exact figure.
- Model predictions are estimates based on historical patterns and may not hold if underlying business conditions change. They should inform, not replace, domain expertise.

11. Appendix

Full model metrics

Metric	Value
Accuracy	0.6667
Precision	0.6675
Recall	0.6667
F1 Score	0.667
ROC-AUC	0.7157
Cv Mean	0.6485
Is Weak	False

Preprocessing

Setting	Value
Test Size	0.2
Cv Folds	5
Random seed	42
Model version	4

Hyperparameters

Parameter	Value
Random Seed	42

Dataset statistics

Statistic	Value
Rows	3,000
Columns	22